

Contactless Vision-based Pulse Rate Detection of Infants under Neurological Examinations

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Abstract—In this paper, we propose a method for detecting variations in the Pulse Rate (PR) of infants undergoing the Hammersmith Infant Neurological Examinations (HINE) using video data. As in every other medical examination the measurement of the PR is critical to underpin the physiological state of living beings. During HINE, measuring the infant's PR is important as its variations against physical conditions, age and other factors must be studied and correlated against developmental scores. However, this becomes highly complicated with active infants where their movements often lead to inconsistent PR estimation. We propose the use of a non-linear dimensionality reduction technique, called Laplacian Eigenmap (LE), to uncover the pulse information encapsulated within the high dimensional visual manifold characterized by normalized RGB feature vectors. Furthermore, low-level image filtering is applied to accurately detect PR within a chosen region-of-interest (ROI) from different parts of the infant's body. For validation and analysis, a set of 14 video sequences of infants undergoing five important tests of HINE have been chosen. Experimental results suggest that a bi-parametrized combination of color features from the RG and GB channels provide more valuable information in comparison to the RB and RGB channels. Results have demonstrated that this contactless method of PR detection has promising prospects for its future use in other clinical examinations of infants.

I. INTRODUCTION

Sensor based measurement of PR from an infant undergoing HINE [1] is often complicated due to the instability of the sensor attached to the infant which is triggered by erratic movements of the infant during examination. Such instability can lead to inconsistent PR measurements and hence cause inconvenience to the medical examiner. However, with the recent developments in computer vision, constant monitoring of variations in the PR of infant using contact-less approach have proven to be quite useful over the conventional sensor-based approach. In addition to having a positive impact on the medical examination process, the procedure could be a pleasant and less fearful experience to the infants undergoing examinations such as the HINE.

Contactless PR measurement from an infant is based on the concept of photo plethysmography (PPG) [2], [3]. According to the principle of PPG, the amount of light incident on the skin and underlying veins has sufficient depth in penetration to pick any fluctuations in blood flow across the vessels

under skin. Since the amount of light reflected from the skin surface is a function of this volumetric change in the blood under the skin, blood volume pulse (BVP) signal can be extracted by analysing the change in skin color of the subject using videos recorded during examinations. Several recent research methodologies have adopted this concept to extract BVP signal from facial regions [4]–[8]. While these exiting methods have focused on the extraction of BVP signal, instead, our method focuses on determining the changes in the PR during conventional medical examinations. In addition, our method is among the few that provides pose independent PR measurement on infants in a manner that is contactless thereby guaranteeing both comfort and accuracy during neurological examinations.

Majority of the above mentioned methods rely on state-of-the-art signal processing techniques such as principle component analysis (PCA), independent component analysis (ICA), linear discriminant analysis (LDA), Laplacian eigenmap (LE) [8], ISOMAP [9], local linear embedding (LLE) [10], local tangent space alignment (LTSA) [11], maximum variance unfolding (MVU) [12], and linearity preserving projection (LPP) [13]. These techniques can be applied on raw signals for extracting the hidden pulse frequency from a high dimensional manifold characterized by the linear and non-linear combination of color features from images and videos. Wei et al. [8] has shown that the manifold learning method of LE can demonstrate superior characteristics as compared to other methods. For example, LE is a non-linear dimensionality reduction method, which does not assume linear combination of features during extraction of pulse information usually adopted in PCA, ICA or LDA-based techniques. In comparison to other non-linear dimensionality reduction methods such as LLE, LTSA or LPP, LE is computationally inexpensive [8]. Furthermore, other non-linear methods such as ISOMAP and MVU are less accurate due to algorithmic limitations and complexity [8]. Finally, LE features are less affected by changes in lighting caused by motion of the subject under investigation as compared to other methods. Therefore, in this paper, LE features are chosen to be extracted for PR estimation during HINE tests. In this paper, we present a systematic study of PR measurement in infants during the course of HINE tests within a typical laboratory environment. We aim to show that the proposed method can be used to estimate the variations in PR during various tests under HINE. Thus, our main scientific objective is to determine correlations between subsequent PR measurements obtained from arbitrary positions of an infant undergoing HINE using a computer vision pipeline

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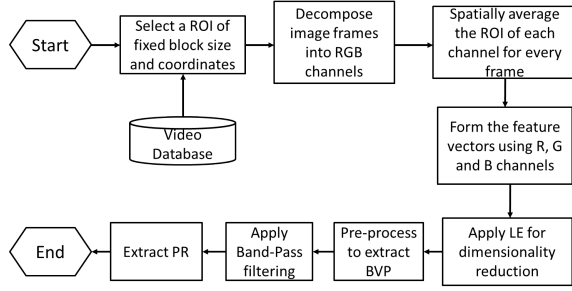


Fig. 1. A block diagram representing the vision-based pipeline used to estimate the pulse rate of infants undergoing HINE.

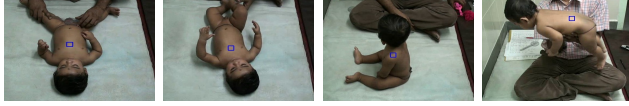


Fig. 2. Sample images of an infant undergoing HINE showing region-of-interest (ROI) marked over the body during adductors, supine-body, trunk-in-sitting, and ventral-suspension examination (left-to-right).

for detecting PR variations over time. The rest of the paper is organized as follows: In Section II, we present the proposed methodology for PR estimation using LE-based color features. Experimental results on video sequences captured during typical HINE tests are presented in Section III. Finally, conclusions leading to future work are described in Section IV.

II. PROPOSED METHODOLOGY

In this section, we describe the vision-based pipeline for PR estimation in infants undergoing HINE tests. Fig. 1 illustrates the proposed framework for PR estimation.

Region-of-Interest (ROI) Selection: The first step in the proposed vision-based pipeline is the ROI selection. A rectangular bounding box enclosing a specific region of the infant's body is initially selected manually for further visual interpretation and understanding. The coordinates of the selected ROI is fixed for all subsequent frames of the video for consistent measurement of the PR undergoing that respective examination. In Fig. 2, four sample frames illustrating the process of ROI selection during adductors-angle-measurement, supine-body, trunk-in-sitting, and ventral-suspension examinations are presented.

Feature Selection through Image Decomposition: Further to selecting an appropriate ROI, it is critical to extract low-level visual features that can be subjected to visual interpretation for PR estimation. Our brief survey of the literature has indicated that the use of color features is typical for this purpose [4]. Here, we decompose the ROI in the image using bi-parametrized color channels, i.e., exploring features in the RG, GB, RB and the joint RGB color spaces. Next to decomposition, all pixels are spatially averaged across each channel (groups in the case of RG, GB and RB). These averaged pixel values are computed over subsequent frames forming independent traces of pixel intensities over time. Therefore, for the i^{th} frame, color feature vector formed

from feature selection through image decomposition can be expressed using (1), where, $c = 1, 2, 3$ for three color channels and x_i^c corresponds to the average pixel value of the c^{th} channel in i^{th} frame. Each of these color feature vectors X_i^c for all the n frames of the chosen video sequence are then fed into the LE signal decomposition algorithm for extracting the raw BVP.

$$X_i^c = [x_1^c, x_2^c, \dots, x_n^c]. \quad (1)$$

LE-based BVP Estimation: As aforementioned, LE is a non-linear dimensionality reduction technique that aims at mapping each data point from a higher dimensional space to a lower dimension based on a specific optimization criterion. According to the Beer-Lambert law, light transmitted or reflected across a tissue region varies non-linearly with distance. The reflected signal, captured within a number of image frames containing the BVP after absorption at the blood cells, is not sufficient to accurately measure BVP, thus making LE one of the most preferred choices for estimating BVP. Here, the feature vector X_i^3 is mapped into one-dimensional space Y_i using the LE technique as outlined in Algorithm 1.

Algorithm 1 LE Algorithm [8]

- 1: **Step 1:** Construction of the adjacency graph G using k-NN;
 - 2: Initialize $k = 12$;
 - 3: **for** $i = 1; i < n; i++$ **do**
 - 4: **if** x_j is one of the k most recent points of x_i **then**
 - 5: x_i and x_j are assumed to be adjacent;
 - 6: **end if**
 - 7: **end for**
 - 8: **Step 2:** Computation of the weight matrix W :
 - 9: **if** x_i and x_j are adjacent **then**
 - 10: $w_{ij} = 1$;
 - 11: **else**
 - 12: $w_{ij} = 0$;
 - 13: **end if**
 - 14: **Step 3:** Divide G into m connected components, e.g. $G = \{G_1, G_2, \dots, G_m\}$.
 - 15: **for** $r = 1; r < m; r++$ **do**
 - 16: $D_{rr} = \sum_r w_{ij}$;
 - 17: **end for**
 - 18: Compute Laplacian matrix L such that $L = D - W$;
 - 19: **Step 4:** Compute of Low dimensional embedding $Y = [Y_1, Y_2, \dots, Y_m]$ by solving $LY = \lambda DY$;
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Y_i estimated using the above algorithm represents the raw BVP signal which can be further processed to extract the PR. Thus, the main objective of LE technique can be represented as a minimization problem as given in (2). This signifies, if x_i and x_j are close to each other in the original signal, then the corresponding Y_i and Y_j in the lower dimensional space are also expected to be close to each other.

TABLE I

TABLE SHOWING KS-TEST STATISTICS OVER FIVE HINE EXPERIMENTS

Pair-wise experiments	KS-Test for RGB p-value	KS-Test for RG p-value	KS-Test for GB p-value	KS-Test for RB p-value
1-2	0.5407	0.1105	0.8621	0.1105
1-3	0.2672	0.2672	0.5407	0.5407
1-4	0.2672	0.1105	0.5407	0.0029<0.05
1-5	0.2672	0.0388<0.05	0.2672	0.2672
2-3	0.8621	0.5407	0.5407	0.5407
2-4	0.5407	0.5407	0.1105	0.0388<0.05
2-5	0.8621	0.8621	0.5407	0.2672
3-4	0.8621	0.8621	0.997	0.0388<0.05
3-5	0.8621	0.2672	0.2672	0.5407
4-5	0.2142	0.5407	0.0388<0.05	0.2672

Note: 1-adductors; 2-Supine Body; 3-Trunk in Sitting; 4-Ventral Suspension; 5-Vertical Suspension

$$\sum_{i,j} \|Y_i - Y_j\|^2 w_{ij}. \quad (2)$$

A. Extraction of Pulse Rate from BVP

The raw BVP pulse extracted using LE is subjected through a five point moving average filter after removing singular points to suppress sharp noise components. Points that are greater than 10 times of the average value of the signal are considered to be singular points. The signal thus obtained is passed through a Finite Impulse Response (FIR) band pass filter using a 15-point Hamming window within 1.16-4 Hz cut-off frequency. A cubical spline interpolation is then applied to ensure the existence of continuous first-order differential. Finally, a peak detection algorithm is used to detect the first order differential characteristic points to demarcate each cycle of the pulse interval. The average time interval is then computed using (3), where, z is the total number of demarcated characteristics points.

$$\bar{T} = \frac{\sum_{s=2}^z (t_s - t_{s-1})}{s-1}. \quad (3)$$

If the interval time is greater than one second, it is considered as two-time intervals and if it is less than three seconds then it is merged with the next interval. Hence the pulse frequency can be calculated using (4).

$$f_{HR} = 60/\bar{T}. \quad (4)$$

III. RESULTS

In this section, we present experimental results validating the proposed method on 14 different video sequences. These videos were recorded at the Neuro-development clinic of The Institute of Post Graduate Medical Education and Research (IPGMER), and SSKM Hospital, Kolkata in INDIA during the period between 2010 to 2011. In a conventional HINE, the infant can be subjected to upto 32 tests that are usually classified into 3 major groups. However, for the purposes of our experiments, we have chosen a subset of 5 tests including adductors-angle-measurement, supine-body, trunk-in-sitting, ventral-suspension, and vertical-suspension. Across these chosen test, we have analysed video sequences of 14 infants within the age group of 3-15 months. We have manually marked ROI at maximally stable and sufficiently exposed

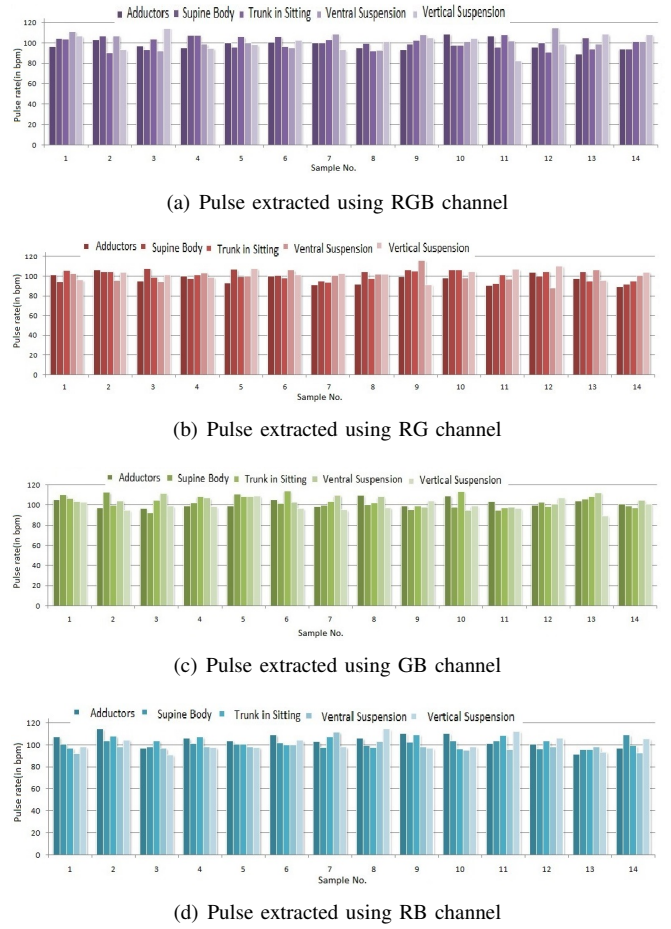


Fig. 3. Pulse rate extracted from the infants during HINE using all possible combinations of RGB color channels

regions in infant's body to extract the PR. It is important to note that the coordinates of the ROI marked for each HINE test remains fixed across all frame of the video sequences and for different tests, different locations of the ROI are chosen. We have studied the effect of varying combinations of color channels, e.g. RG, GB, RB and RGB to support our analysis. Results of such experiments are presented systematically as below.

We begin by performing the Kolmogorov-Smirnov (KS) test to measure the similarity between two sets of measurements over all samples assuming a null hypothesis that all experiments belong to the same distribution, i.e. there exists no differences between the measurements made from different color space combinations. The KS-test is carried over several combinations of color channels and reported in the Table I. Here, the p-value is used as a statistical measure for testing the statistical hypothesis. A threshold value α , also called the significance level, is set prior to experimentation. If p-value is found equal or less than α , then the measured data is said to be inconsistent with the assumed hypothesis. Traditionally, the significance level is bounded within the range of 5% and hence α is set to 0.05 as the p-value lies within the range of 0 and 1.

TABLE II

TABLE SHOWING MEAN $\pm$ STANDARD DEVIATION OF THE MEASURED PR OVER DIFFERENT EXPERIMENTS USING VARYING CHANNEL COMBINATIONS. PR MEASUREMENT IN **BOLD** INDICATE THE CHANNEL COMBINATION WITH THE LEAST STANDARD DEVIATION

Sample No	RGB	RG	GB	RB
1	104.5 $\pm$ 4.67	100.3 $\pm$ 4.24	105.1 $\pm$ 2.63	99.1 $\pm$ 4.82
2	100.0 $\pm$ 6.96	102.9 $\pm$ 3.81	101.2 $\pm$ 6.25	105.9 $\pm$ 5.49
3	100.4 $\pm$ 8.35	99.5 $\pm$ 4.98	100.2 $\pm$ 6.64	97.4 $\pm$ 4.04
4	100.7 $\pm$ 5.71	100.3 $\pm$ 2.05	102.5 $\pm$ 4.16	102.0 $\pm$ 3.93
5	99.9 $\pm$ 3.45	101.7 $\pm$ 5.46	106.6 $\pm$ 4.12	100.2 $\pm$ 2.23
6	100.1 $\pm$ 3.90	101.4 $\pm$ 2.84	103.6 $\pm$ 5.79	103.2 $\pm$ 3.38
7	101.1 $\pm$ 5.03	96.8 $\pm$ 4.14	100.8 $\pm$ 4.95	103.6 $\pm$ 5.38
8	96.2 $\pm$ 3.67	99.7 $\pm$ 4.30	103.1 $\pm$ 4.71	104.0 $\pm$ 5.98
9	101.4 $\pm$ 5.08	104.0 $\pm$ 8.07	98.7 $\pm$ 2.81	103.6 $\pm$ 5.51
10	101.9 $\pm$ 4.16	103.0 $\pm$ 3.81	102.5 $\pm$ 7.15	100.8 $\pm$ 5.82
11	99.0 $\pm$ 9.40	97.9 $\pm$ 5.83	97.5 $\pm$ 3.07	104.5 $\pm$ 5.72
12	100.1 $\pm$ 8.05	101.7 $\pm$ 7.28	101.3 $\pm$ 3.11	101.0 $\pm$ 3.41
13	99.1 $\pm$ 7.06	100.1 $\pm$ 4.59	103.5 $\pm$ 7.77	95.1 $\pm$ 2.31
14	99.9 $\pm$ 5.21	96.3 $\pm$ 5.23	100.2 $\pm$ 2.57	100.7 $\pm$ 5.90

From Table I, it can be deduced that the KS-test performed on the estimated PR data using the RGB combined channel is generally in accordance to the null hypothesis. However, different combinations of channels in a pairwise manner exhibit some level of inconsistency that can particularly be noticed in the case of RG and GB for the pairs 1-5 and 4-5 respectively. Similarly, in the case of the RB combination, three pairs: 1-4, 2-4 and 3-4 are inconsistent against the null hypothesis. Given the extent of inconsistency, it is acceptable to infer that amongst all combination of channels, the RB is the least reliable combination for PR estimation. This result is in agreement with the previous proposition that the green channel mostly carries the BVP pulse signal measured using the PPG technique as mentioned in [4].

In Table II, we present the mean of the estimated PR across all five HINE tests covering the 14 infants and deviation of the PR from the mean across all combinations of color channels. From Table II, it can be noted that the channels RG and GB have the least deviation for atleast 5 infants across all experiments. Therefore, it can be stated that, both RG and GB channel combinations are most stable amongst all combinations. On the other hand, RGB and RB provide least deviation for infants 1 and 3 respectively. Hence, RGB and RB can be considered to be less consistent as compared to RG and GB channels.

Next, we present measurements of the PR for all samples. In Fig. 3, the four plots illustrate the extent of closeness of the PR for each infant over different HINE tests. Figs. 3 (a-d) represent variations of PR during five experiments using various color channel combinations. From Fig. 3, it may be observed that, the PR estimated is quite consistent for a majority of the samples when the RG and GB channel combinations are used. The average computational time using the RGB channels together has been found to be 4.945ms, whereas, while using the RG, GB and RB channel combinations it takes 4.563ms, 7.790ms and 5.922ms, respectively. Finally, it can be highlighted that the range of PR estimated using the proposed approach lies within the actual range of 85 bpm to 120 bpm for infants.

IV. CONCLUSION

This paper proposes a feasibility study on the application of contactless PR estimation during neurological exami-

nation of infants. This study has explored the effect of bi-parametrization of the RGB color space and its pairwise effect on PR estimation of infant undergoing HINE. Experimental results on several samples have demonstrated that the RG and GB channel combinations are more accurate in comparison to the RGB or RB channel combinations. Both our statistical tests and empirical experiments on the estimated PR provide consistent analysis, and the possibility of inferring that the G channel is among the most important for PR extraction as it carries the BVP signal.

We have found that the proposed method cannot cope with rapid movements of the infant during the examination or under drastically changing illumination conditions. Despite all limitations, given a stable laboratory environment for testing, PR of infants undergoing HINE can be estimated and correlated with their developmental scores successfully. This study can also be considered as the first of the contactless approaches for PR measurement for infants who are highly susceptible to injuries from instruments used in other contact-based methods. In addition, we expect that the method can also be extended for the extraction of other vital signals, analysed for other HINE test and a longitudinal study can be carried out for further investigation.

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